MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3440 HOT ACID CURE
DESIGN OF TANK FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to Prepare for design of **HOT ACID CURE (3440-REA-001 & 002)-** Reactor Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel . For member strength calculation force values are directly taken from Load Data Provided.

4. DESIGN REQUIREMENTS

4.1 Design Loads

All Design Loads are indicated in 3440-REA-001,002 HOT ACID CURE REACTOR (Doc. No. MD-216-3440-EG-ME-PLN- 1001) attached in Attachment 1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

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5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Aboveground Tanks

Foundation shall be Earth Foundation with a Concrete Ringwall for Tanks. The design calculations are prepared as per recommendations of standard API 650 - Annex B.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	50

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 350 kN/m^2 at 1.25m depth below EGL (ie. RL 911.650) refer Table 5 - BH-9 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

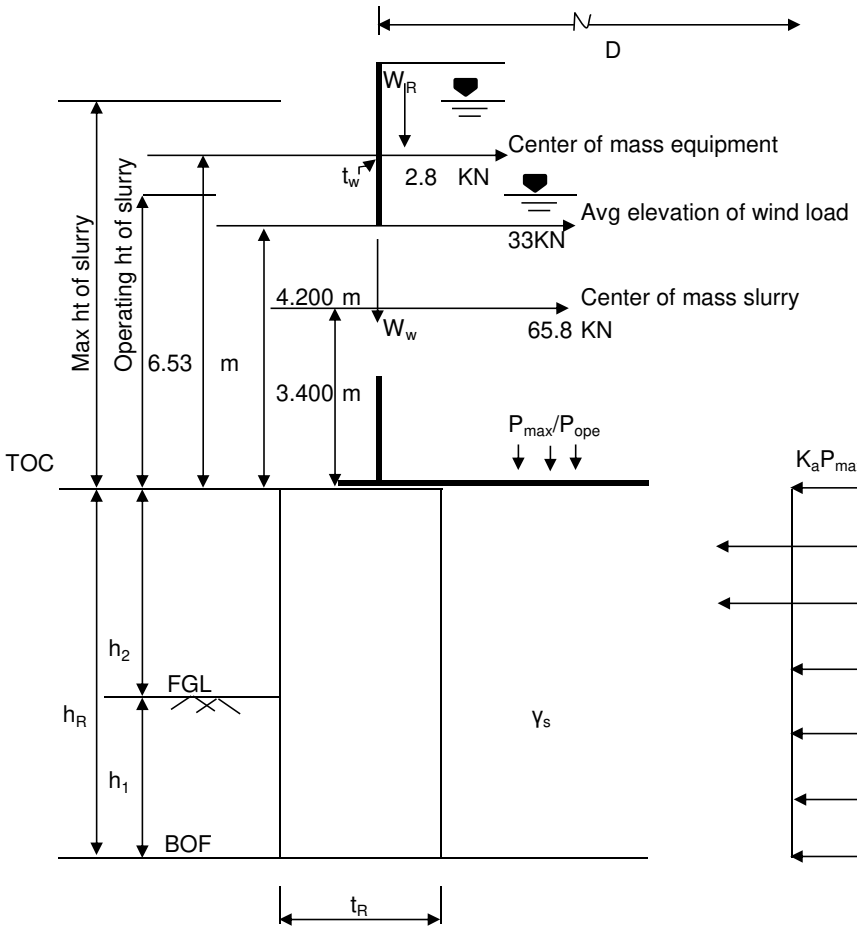
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6. DESIGN CALCULATION FOR FOUNDATION

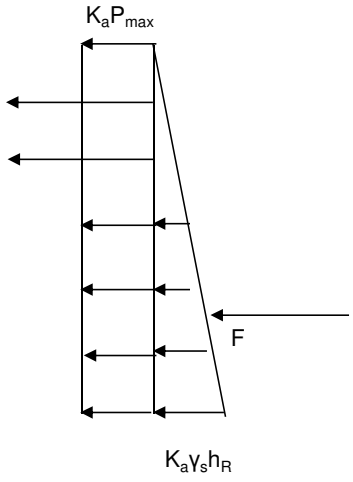
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Design Calculation of Foundation for HOT ACID CURE REACTOR (3440-REA-001 & 002)



Typical Section Through Tank



Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	5.0	m
Self weight of tank =	113.00	KN
Self weight of tank per m of circumference,W_w =	7.194	KN/m
External platform weight =	49	KN
Self weight of tank roof per m of circumference,W_R=	3.119	KN/m
Maximum weight of slurry=	1674	KN
Pressure at base of tank ,P_{max}	85.26	KN/m^2
Operating weight of slurry=	1387	KN
Pressure at base of tank ,P_{ope}	70.64	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	350	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.25	m	
FGL to Bottom of foundation ,h_1 =	1.25	m	
FGL to Top of concrete ,h_2 =	1	m	(h2 value is adopted from 3440-REA-001
Ringwall thickness, t_R =	450	mm	to be on the conservative side)
Inner dia of ringwall,D_i =	4.55	m	
Outer dia of ringwall,D_o =	5.45	m	
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	8.17	m^3	

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 33 kn acting at a height of 6.450 m from BOF

M_{OT} = Overturning moment = 212.9 KN-m

M_{OT} / S_{xx} = 26 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 120 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 146 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 93.5 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 76.9 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 103.0 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 50.87 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 212.9 KN-m

M_{OR} = Restoring moment = (113+PI()*5*0.45*2.25*25)*5/2 = 1276.5 KN-m

FOS = M_{OR} / M_{OT} 6.00 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Moment at base level due to seismic force = 242 Kn-m

Seismic force at center of mass equipment = (113/(113+1387)*242)/6.53 = 2.792 KN

acting at a height of 8.780 m from BOF

Seismic force at center of mass slurry = (1387/(113+1387)*242) / 3.4 = 65.8 KN

acting at a height of 5.650 m from BOF

M_{OT} = Overturning moment = 396 KN-m

M_{OT} / S_{xx} = 48.5 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 112 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 146 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 78.3 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + 0.61 \times P_{max} + h_R\gamma_c =$ 128.9 Kn/m^2 < 350.0 KN/m^2 Ok

As the tank encroaches the ring wall by 275mm out of total 450mm , the pressure under the tank is reduced to 275/450 = 0.61 times.

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s =$ 128.0 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2) =$ 79.17 Kn/m

Hoop tension in ringwall, $T = 1/2FD =$ 198 Kn

Factored design tension, $T_u =$ 277 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y =$ 733.1 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R =$ 2025 mm^2

$A_{s,reqd}$ (per face of ring wall)= 1013 mm^2

Required : 12 dia bar @ 251.5 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 1273 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R =$ 540 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291.1 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 392.9 mm^2 ok

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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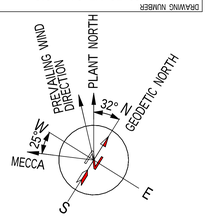
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ATTACHMENT 1 INPUT DRAWINGS

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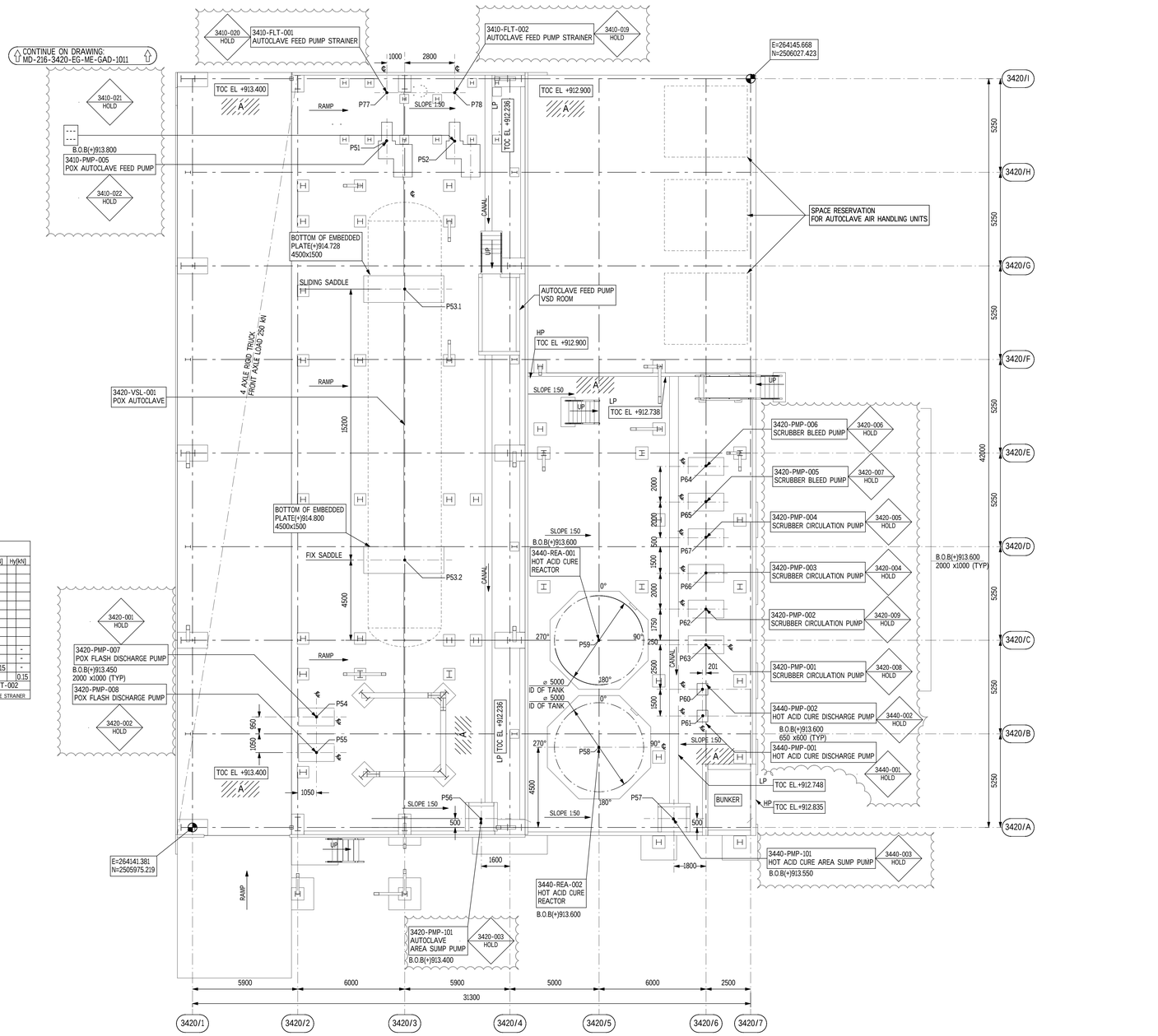


*NOTE:
FOR LOAD P58 & P59, REFER DRAWING NUMBER MD-216-3440-EG-ME-PLN-1001
FOR LOAD P53, REFER DRAWING NUMBER MD-216-3420-EG-ME-PLN-1001

LOADING POINT	P51				P52				P53.1/53.2 *				P54				P55				P56			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT	- 130				- 130								- 7				- 7				- 10			
2 DEAD LOAD (SECONDARY EQUIPMENT)																								
3 OPERATION LOAD 1. FILLING																								
4 OPERATION LOAD 2. PRESSURE																								
5 OPERATION LOAD 3. TEMPERATURE																								
6 OPERATION LOAD 4. DYNAMIC	- 195				- 195								- 11				- 11				- 15			
7 OPERATION LOAD 5. OTHERS																								
8 TEST LOAD FILLING / EMERGENCY																								
9 LIVE LOAD																								
10 WIND ±X - DIRECTION	-	-	-		-	-	-						-	-	-		-	-	-		-	-	-	
11 WIND ±Y - DIRECTION	-	-	-		-	-	-						-	-	-		-	-	-		-	-	-	
12 EARTHQUAKE ±X - DIRECTION	-	4.5	-		-	4.5	-						-	0.35	-		-	0.35	-		-	0.5	-	
13 EARTHQUAKE ±Y - DIRECTION	-	-	4.5		-	-	4.5						-	-	0.35		-	-	0.35		-	-	0.5	
EQUIPMENT - NO.	3410-PMP-005				3410-PMP-006				3420-VSL-001				3420-PMP-007				3420-PMP-008				3420-PMP-101			
DESCRIPTION	AUTOCLAVE FEED PUMP				AUTOCLAVE FEED PUMP				POX AUTOCLAVE				POX FLASH DIS. PUMP				POX FLASH DIS. PUMP				AUTOCLAVE SUMP PUMP			

LOADING POINT	P57				P58*				P59*				P60				P61				P62			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]	
1 DEAD LOAD EQUIPMENT	- 8												- 6				- 6				- 10			
2 DEAD LOAD (SECONDARY EQUIPMENT)																								
3 OPERATION LOAD 1. FILLING																								
4 OPERATION LOAD 2. PRESSURE																								
5 OPERATION LOAD 3. TEMPERATURE																								
6 OPERATION LOAD 4. DYNAMIC	- 12												- 9				- 9				- 15			
7 OPERATION LOAD 5. OTHERS																								
8 TEST LOAD FILLING / EMERGENCY																								
9 LIVE LOAD																								
10 WIND ±X - DIRECTION	-	0.5	-										-	0.5	-		-	0.5	-		-	0.5	-	
11 WIND ±Y - DIRECTION	-	-	0.5										-	-	1.0		-	-	1.0		-	-	1.0	
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DESCRIPTION	HOT ACID SUMP PUMP				HOT ACID REACTOR				HOT ACID REACTOR				HOT ACID DIS. PUMP				HOT ACID DIS. PUMP				SCRUBBER CIRC. PUMP			

LOADING POINT	P63				P64				P65				P66				P67				P77				P78			
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]	Hy[kN]		Vz[kN]	Hx[kN]		Hy[kN]	Vz[kN]	Hx[kN]		Hy[kN]
1 DEAD LOAD EQUIPMENT	- 10				- 5				- 5				- 10				- 10				- 3				- 3			
2 DEAD LOAD (SECONDARY EQUIPMENT)																												
3 OPERATION LOAD 1. FILLING																												
4 OPERATION LOAD 2. PRESSURE																												
5 OPERATION LOAD 3. TEMPERATURE																												
6 OPERATION LOAD 4. DYNAMIC	- 15				- 7.5				- 7.5				- 15				- 15											
7 OPERATION LOAD 5. OTHERS																												
8 TEST LOAD FILLING / EMERGENCY																												
9 LIVE LOAD																					- 5				- 5			
10 WIND ±X - DIRECTION	-	0.5	-		-	0.5	-		-	0.5	-		-	-	-		-	-	-		-	-	-		-	-	-	
11 WIND ±Y - DIRECTION	-	-	1.0		-	-	1.0		-	-	1.0		-	-	-		-	-	-		-	-	-		-	-	-	
12 EARTHQUAKE ±X - DIRECTION	-	0.5	-		-	0.1	-		-	0.1	-		-	0.5	-		-	0.5	-		-	0.15	-		-	0.15	-	
13 EARTHQUAKE ±Y - DIRECTION	-	-	0.5		-	-	0.1		-	-	0.1		-	-	0.5		-	-	0.5		-	-	0.15		-	-	0.15	
EQUIPMENT - NO.	3420-PMP-001				3420-PMP-006				3420-PMP-005				3420-PMP-003				3420-PMP-004				3410-FLT-001				3410-FLT-002			
DESCRIPTION	SCRUBBER CIRC. PUMP				SCRUBBER BLEED PUMP				SCRUBBER BLEED PUMP				SCRUBBER CIRC. PUMP				SCRUBBER CIRC. PUMP				AUTOCLAVE STRAINER				AUTOCLAVE STRAINER			



PLAN AT EL. (+)912.900

0 1 10m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
CHECKERED PLATE
GRATING
GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS: +Z, +Mz, +Y, +X, +My, +Mx

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
T.O.R. = TOP OF RAIL
OPERATING PLATFORM- GRATING 5 kN/m²
GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN.

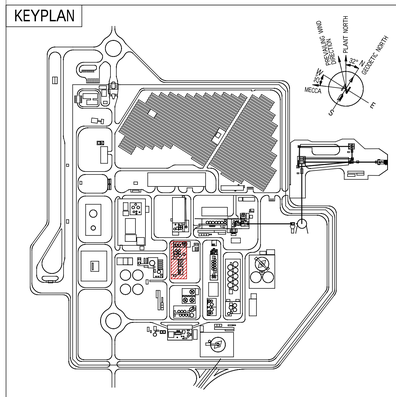
13 - REPORTED SEISMIC LOAD ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

14 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.900 m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3420-EG-ME-GAD-1015	LOAD PLAN, AUTOCLAVE, PLAN AT EL. 929.000
MD-216-3420-EG-ME-GAD-1014	LOAD PLAN, AUTOCLAVE, PLAN AT EL. 920.100
MD-216-3420-EG-ME-GAD-1012	LOAD PLAN, FEED STORAGE, PLAN AT EL. +929.300
MD-216-3420-EG-ME-GAD-1011	LOAD PLAN, FEED STORAGE, PLAN AT EL. +912.900
MD-216-3420-EG-ME-GAD-1001	GENERAL ARRANGEMENT DRAWING, PLAN

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	06-03-2020	ISSUED FOR DESIGN	MA	TV	SC	RL	
A01	08-01-2020	ISSUED FOR APPROVAL	SF	TV	SC	RL	

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LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_450_M_D_02688_003	1
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707 _GAU01 _HH12 _MLD14 _34203	OU100667823 / 1

DRAWN	SF	08-01-2020
CHECKED	TV	08-01-2020
ENGINEER	SC	08-01-2020
APPROVED	RL	08-01-2020
EMPLOYER		
SCALE	1:100	A0

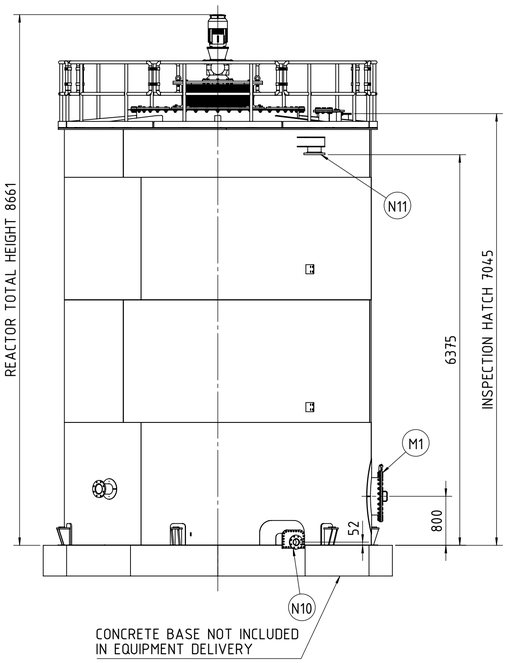
MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
PRESSURE OXIDATION - AUTOCLAVE
PLAN AT EL. +912.900

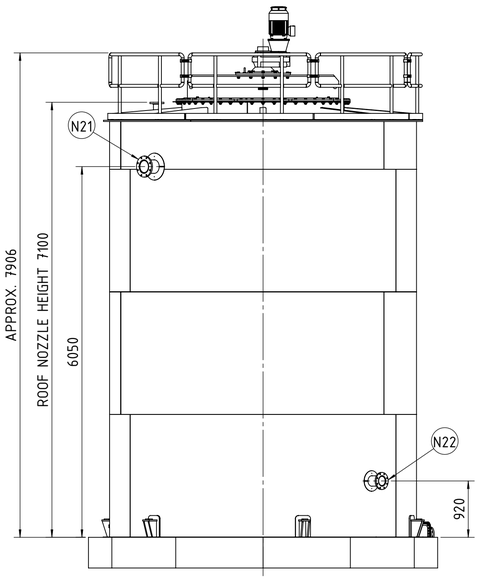
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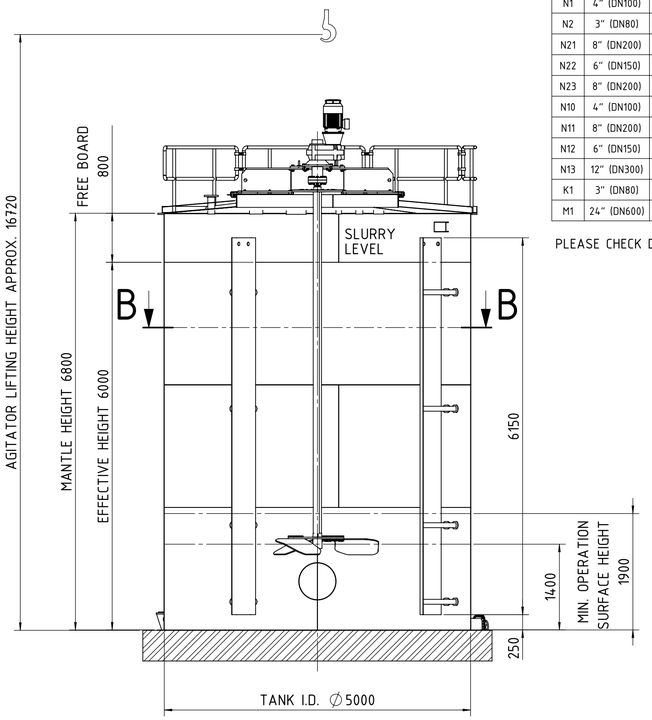
DRAWING NUMBER MD-216-3440-EG-ME-GAD-1001
REV A01

NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N1	4" (DN100)	CLASS 150	20°	7100	ASME B16.5	SLURRY INLET	1	
N2	3" (DN80)	CLASS 150	10°	7100	ASME B16.5	SUMP SLURRY INLET	1	
N21	8" (DN200)	CLASS 150	225°	6050	ASME B16.5	SLURRY OUTLET	1	
N22	6" (DN150)	CLASS 150	135°	920	ASME B16.5	SLURRY OUTLET	1	
N23	8" (DN200)	CLASS 150	330°	7100	ASME B16.5	VENT	1	
N10	4" (DN100)	CLASS 150	65°	52	ASME B16.5	DRAIN	1	
N11	8" (DN200)	CLASS 150	55°	6375	ASME B16.5	EMERGENCY OVERFLOW	1	
N12	6" (DN150)	CLASS 150	30°	7100	ASME B16.5	SPARE	1	WITH BLIND FLANGE
N13	12" (DN300)	–	350°	7045	OUTOTEC	INSPECTION HATCH	1	
K1	3" (DN80)	CLASS 150	300°	7100	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	–	0°	800	OUTOTEC	MANHOLE	1	

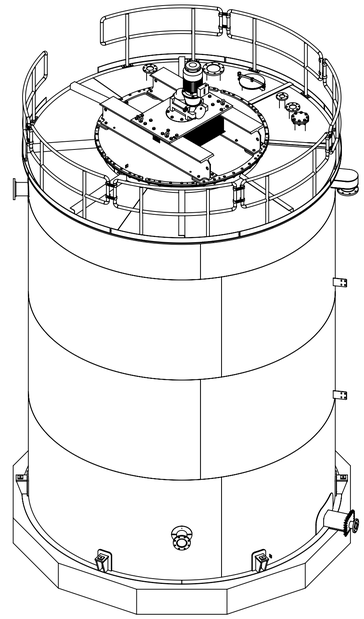
PLEASE CHECK DATASHEET MD-216-3440-EG-ME-DAT-1001 FOR PROCESS VALUES

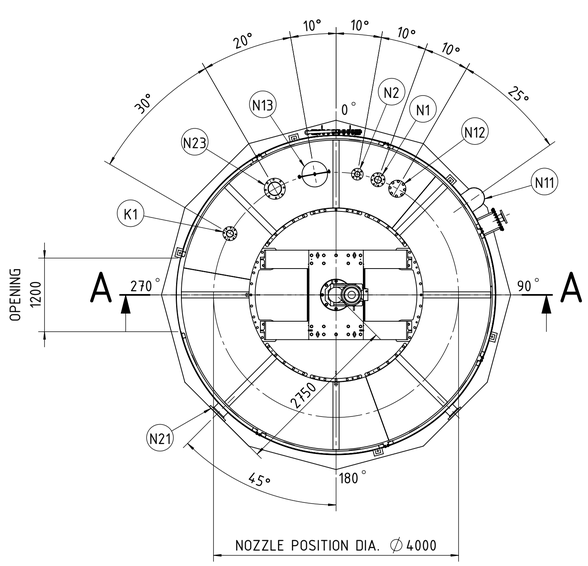


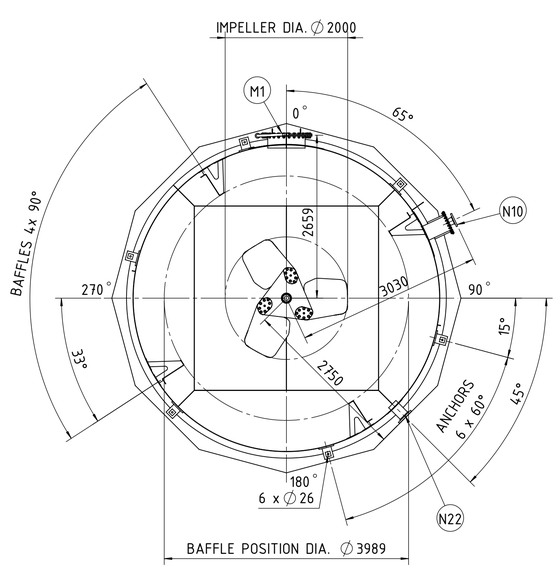




SECTION A-A







SECTION B-B

HOT ACID CURE REACTOR	
AGITATOR WEIGHT APPR. (kg)	2900
DEAD WEIGHT (kg)	11300
OPERATIONAL WEIGHT (kg)	150100
MAXIMUM WEIGHT (kg)	168600

Document Number
MD-216-3440-EG-ME-GAD-1001

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Jan 17, 2020, 9:57 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3440-EG-ME-PLN-1001	FOUNDATION AND LOAD PLAN								
MD-216-3440-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3440-EG-PR-PID-1001	P&I DIAGRAM								
MD-216-3440-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	20.12.2019	ISSUED FOR APPROVAL	AT	TM	JR	GR	

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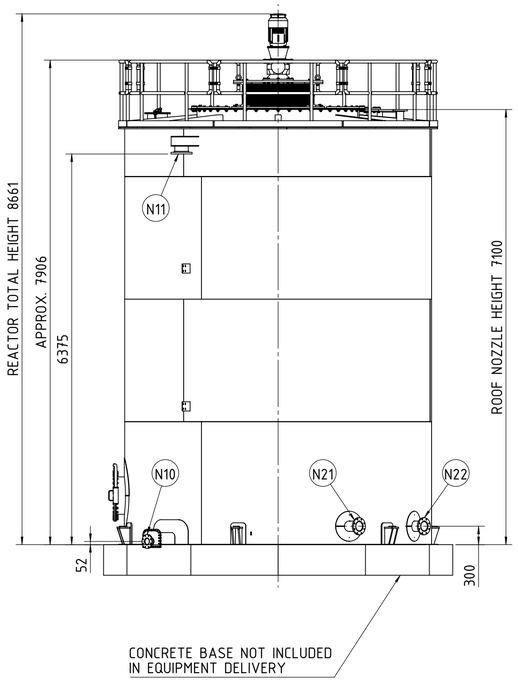
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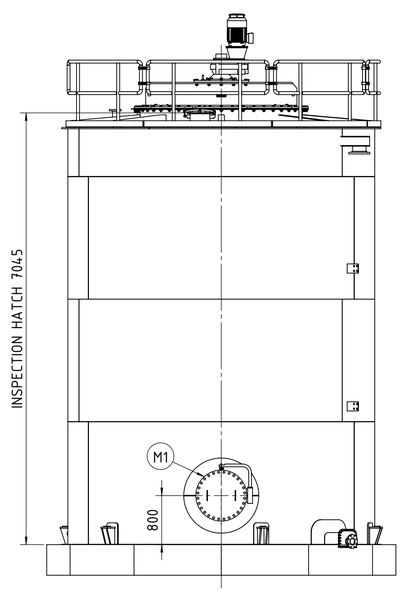
DRAWN	aritik	20.12.2019	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	tuoman	20.12.2019	GENERAL ARRANGEMENT
ENGINEER	juhris	20.12.2019	HOT ACID CURE
APPROVED	GeoRie	20.12.2019	3440-REA-001 HOT ACID CURE REACTOR
EMPLOYER			
SCALE 1:50		A1	DRAWING NUMBER: MD-216-3440-EG-ME-GAD-1001 — REV: A01

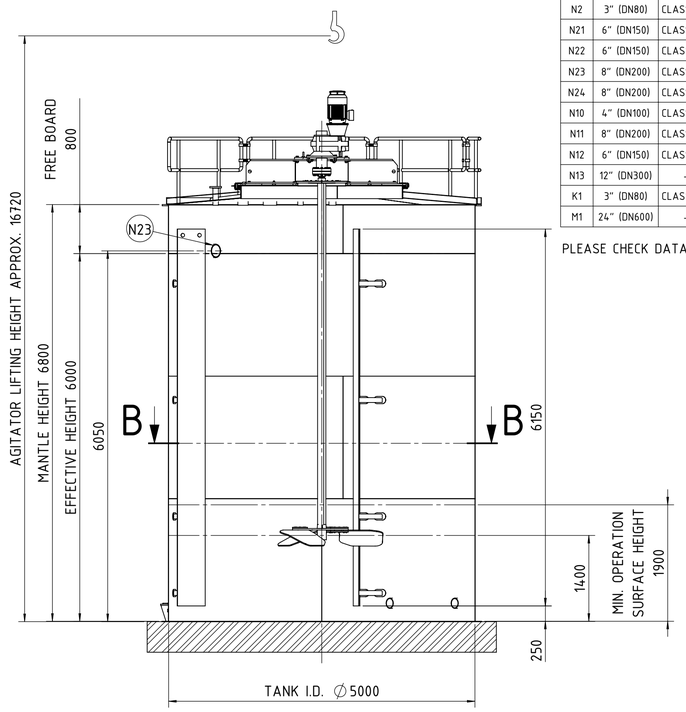
DRAWING NUMBER MD-216-3440-EG-ME-GAD-1002 | REV A01

NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N1	4" (DN100)	CLASS 150	5°	7100	ASME B16.5	SLURRY INLET	1	
N2	3" (DN80)	CLASS 150	350°	7100	ASME B16.5	SUMP SLURRY INLET	1	
N21	6" (DN150)	CLASS 150	45°	300	ASME B16.5	SLURRY OUTLET	1	
N22	6" (DN150)	CLASS 150	45°	300	ASME B16.5	SLURRY OUTLET	1	
N23	8" (DN200)	CLASS 150	315°	6050	ASME B16.5	SLURRY INLET	1	
N24	8" (DN200)	CLASS 150	335°	7100	ASME B16.5	VENT	1	
N10	4" (DN100)	CLASS 150	135°	52	ASME B16.5	DRAIN	1	
N11	8" (DN200)	CLASS 150	125°	6375	ASME B16.5	EMERGENCY OVERFLOW	1	
N12	6" (DN150)	CLASS 150	320°	7100	ASME B16.5	SPARE	1	WITH BLIND FLANGE
N13	12" (DN300)	-	190°	7045	OUTOTEC	INSPECTION HATCH	1	
K1	3" (DN80)	CLASS 150	300°	7100	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	-	180°	800	OUTOTEC	MANHOLE	1	

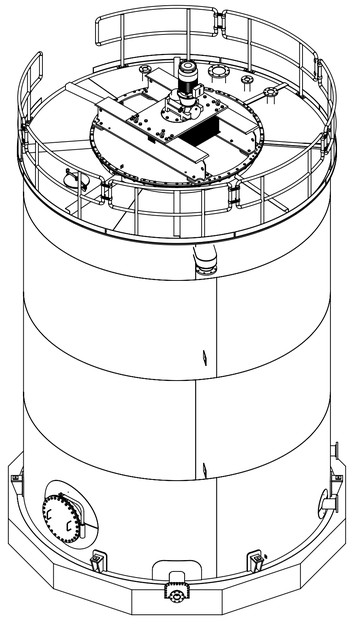
PLEASE CHECK DATASHEET MD-216-3440-EG-ME-DAT-1001 FOR PROCESS VALUES

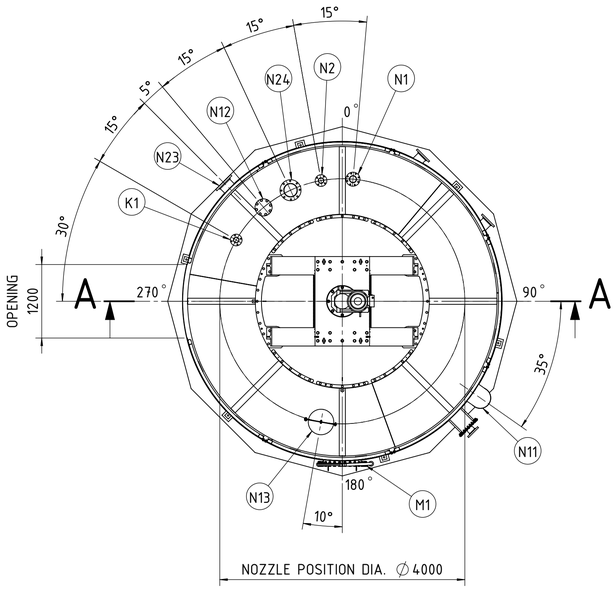


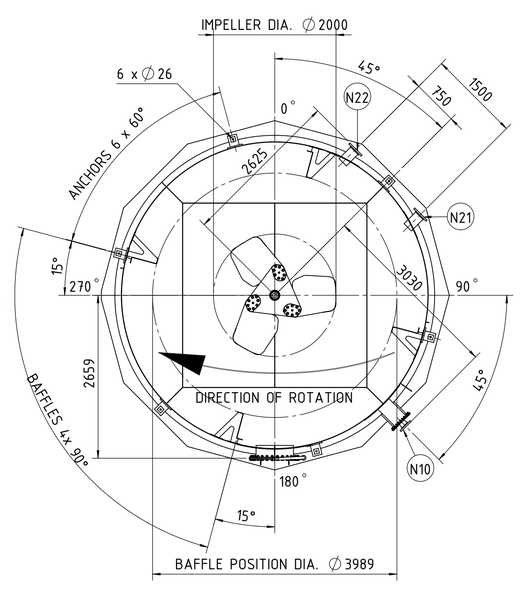




SECTION A-A







SECTION B-B

HOT ACID CURE REACTOR	
AGITATOR WEIGHT APPR. (kg)	2900
DEAD WEIGHT (kg)	11300
OPERATIONAL WEIGHT (kg)	150100
MAXIMUM WEIGHT (kg)	168600

Document Number
MD-216-3440-EG-ME-GAD-1002

Revision A01

Approved

Mr Doron Golan - Bechtel Global

Jan 17, 2020, 9:58 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3440-EG-ME-PLN-1001	FOUNDATION AND LOAD PLAN								
MD-216-3440-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3440-EG-PR-PID-1002	P&I DIAGRAM								
MD-216-3440-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	20.12.2019	ISSUED FOR APPROVAL	AT	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER: O19045-450-M-G-02652-002 | REVISION: 0

OUTOTEC DRAWING NUMBER: 900707 GAU01 RD51 MTL03 34402 | DOCUMENT ID: OU601247725 | REVISION: 0

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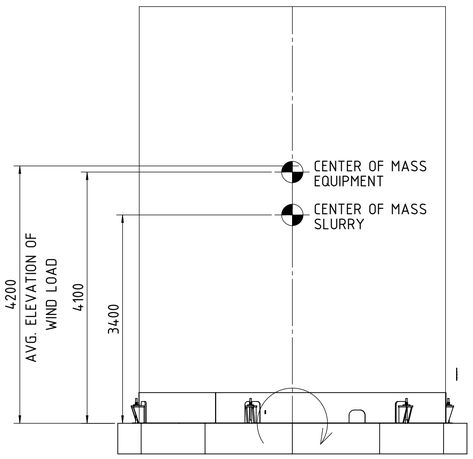
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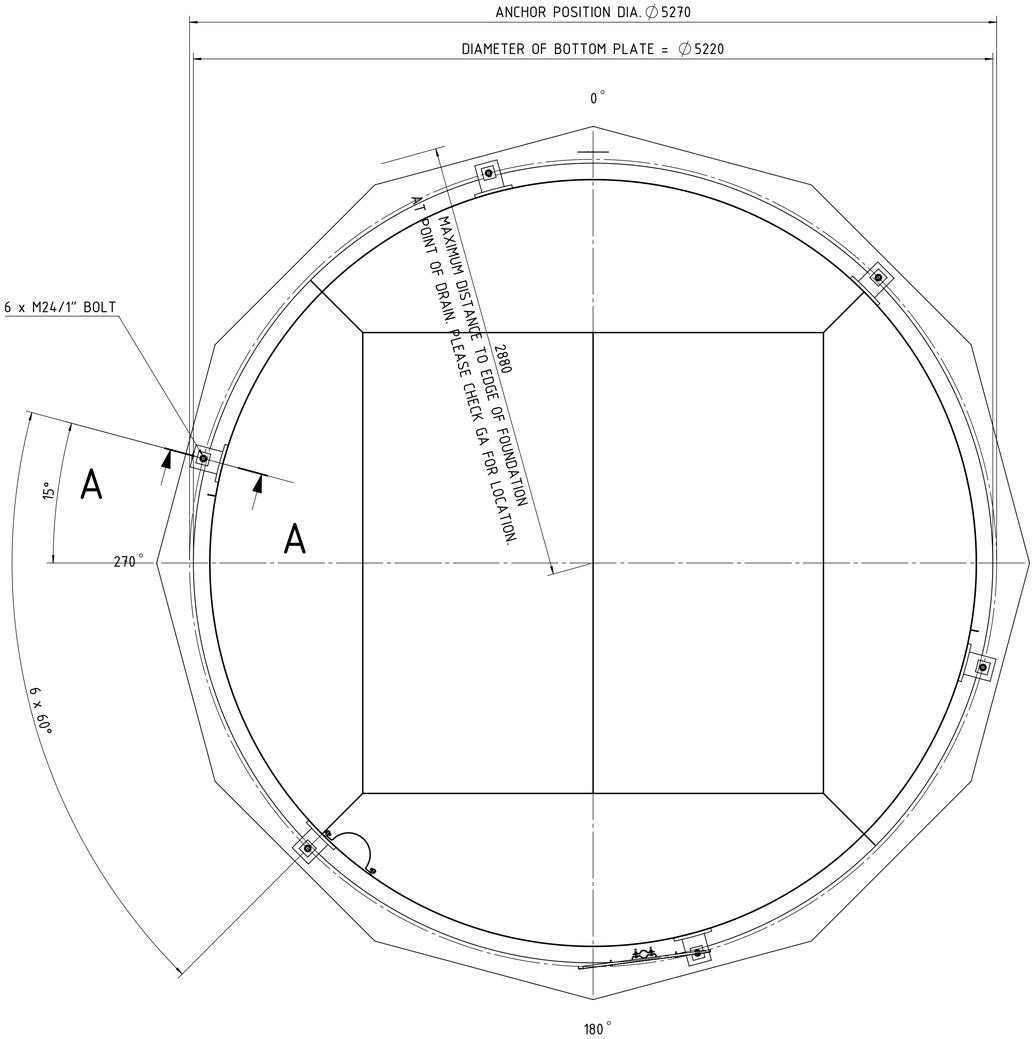
DRAWN	aritik	20.12.2019	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	tuoman	20.12.2019	GENERAL ARRANGEMENT
ENGINEER	juhris	20.12.2019	HOT ACID CURE
APPROVED	GeoRie	20.12.2019	3440-REA-002 HOT ACID CURE REACTOR
EMPLOYER			
SCALE 1:50		A1	DRAWING NUMBER: MD-216-3440-EG-ME-GAD-1002 | REV: A01

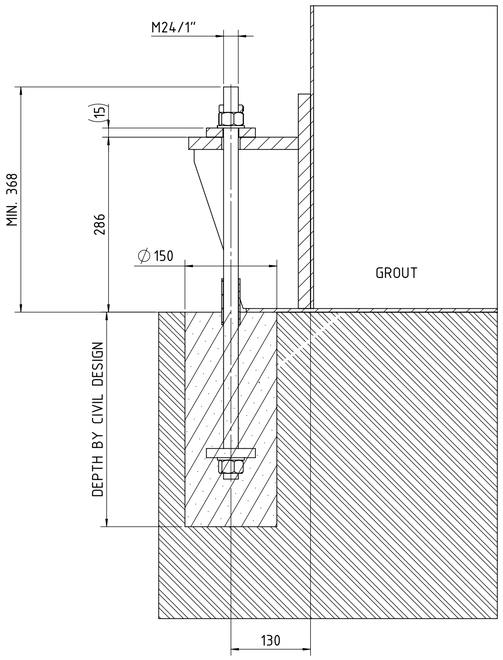
DRAWING NUMBER: MD-216-3440-EG-ME-PLN-1001
REV: D01

LOADING NOTES:	
1.	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2.	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
3.	REPORTED SEISMIC LOADS ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC 301 - 2007. HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNERACCORDING SBC 301 - 2007.
4.	REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC 301 - 2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE SHELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 50





SECTION A-A
SCALE 1 : 5

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. ground elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

HOT ACID CURE TANK	
DEAD WEIGHT	11300 kg
OPERATIONAL WEIGHT OF SLURRY	138700 kg
MAXIMUM WEIGHT OF SLURRY	167400 kg
EXTERNAL PLATFORM LOAD	4900 kg
NORMAL OPERATING SPEED	55,7 rpm
WIND LOAD	33 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	242 kNm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3440-EG-ME-GAD-1001,1002	GENERAL ARRANGEMENT DRAWING								
MD-216-3440-EG-ME-DAT-1001	EQUIPMENT DATASHEET	D01	20.11.2023	APPROVED FOR CONSTRUCTION	VP	TM	JR	GR	
MD-216-3440-EG-PR-PID-1001,1002	P&I DIAGRAM	C01	03.03.2020	APPROVED FOR CONSTRUCTION	VP	TM	JR	GR	
MD-216-3440-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	20.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

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OUTOTEC DRAWING NUMBER: 900707 GAU01 RD51 MLD15 34401 — DOCUMENT ID: OU601266515 — REVISION: 3

DRAWN	vilpoy	27.02.2020
CHECKED	tuoman	03.03.2020
ENGINEER	juhris	03.03.2020
APPROVED	GeoRie	03.03.2020
EMPLOYER		
SCALE	1:20	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
HOT ACID CURE
3440-REA-001, 002 HOT ACID CURE REACTOR

DRAWING NUMBER: MD-216-3440-EG-ME-PLN-1001 — REV: D01